

Vamshika Guthula

United States • +1(973)6107054 • vamshika.guthula@gmail.com • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Versatile Software Engineer with 3+ years of experience in DevOps, cloud infrastructure optimization, system performance enhancement, and end-to-end testing automation. Proficient in delivering scalable cloud solutions, with a proven ability to streamline processes and improve error rates. Committed to innovation and continuous learning to meet business objectives.

EDUCATION

University of Maryland, College Park, MD <i>Master of Science, Telecommunication</i>	GPA 3.4 May 2021
Chaitanya Bharathi Institute of Technology, Hyderabad, India <i>Bachelor of Engineering, Computer Science Engineering</i>	GPA 3.4 May 2018

TECHNICAL SKILLS

Programming: C, C++, Java, Python, Go, Node.js, REST API, JWT, JavaScript, TypeScript, MySQL, Oracle SQL, Shell Scripting
Tools: AWS, ECS, CloudFormation, Azure, Prometheus, Grafana, Git, Postman, Jenkins, Playwright, Go bench, GitHub, GitLab, Docker, Oracle ERP, OBIA, JIRA, ServiceNow, Tableau, MS Office Suite
Operating Systems: Windows, Linux (RHEL), Unix, Mac OS
Certifications: Professional Scrum Master 1 (PSM 1), Oracle Financials, Oracle Cost Management, Oracle Inventory

PROFESSIONAL EXPERIENCE

Origin Wireless Inc., Rockville, MD – Software Engineer Nov 2021 – Jul 2024
Performance measurement and improvements on AWS cloud, coupled with automation of the Home-Awareness (Wi-Fi Motion Sensing and Fall Detection) system

- Developed End-to-End test scripts using simulated device for API testing of servers on Azure Kubernetes Service (AKS) through Kafka using Typescript and Playwright framework enhancing the testing accuracy and reliability
- Containerized test scripts and configured run-time environment variables using Docker decreasing deployment error rate by 8% and server code error rate by 15%
- Updated scripts for performance testing of systems to consolidate metrics like latency and efficiency using Go-Bench to improve system performance by an average of 40%
- Implemented CloudFormation Template (CFT) in YAML for updating AWS server on ECS and database deployment to support about ~2K users per container and programmatically enable auto scaling to support more than 100K users
- Configured Prometheus to gather container metrics to design 7+ dashboards on Grafana representing both CloudWatch and container metrics and, integrated slack to identify triggered notifications improving alert system efficiency by 30%
- Integrated and maintained DevOps CI/CD tools, including Jenkins, Git, and Docker, to rapidly automate delivery pipelines
- Automated processes using Python and Shell scripts by analyzing issues on GitLab to reduce system setup time by 70%

GE Appliances, Hyderabad, India – Associate Systems Analyst (Graduate Engineer Trainee) Jun 2018 – Jul 2019
Served as the IT-Costing Analyst for Supply chain Management from IT- Finance department for kitchen appliances

- Collaborated cross-functionally to develop dashboards to track Stock Keeping Unit (SKU) levels, using Oracle ERP and Oracle Business Intelligence Application (OBIA) to reduce reconciliation time of 7+ systems by 80% through automation
- Migrated data from Oracle 11i to Oracle R12 for Costing and Accounts Receivable across 30K items with over 10K consumers
- Piloted migration of data from previous years of Oracle ERP to archival instances, reducing query time by 30%

Bharat Heavy Electricals Limited (BHEL), Hyderabad, India – Intern in Networking and Security May 2016 – Jun 2016
Observed and built the perimeter level network security at BHEL

- Built a simulation model (3- Pronged Firewall topology) having an internal network, an outside network and a DMZ network using CISCO ASA firewall for perimeter level network security at BHEL

PROJECT EXPERIENCE

Cloud Formation Template on AWS Mar 2020 – Apr 2020

- Designed and deployed a CloudFormation Template using JSON to launch a 3- tiered architecture LAMP (Linux, Apache, MySQL, Perl/PHP/Python) stack
- Utilized AWS CloudFormation Services, AWS Management console, EC2, S3, IAM, RDS, Route 53, ELB and CloudWatch.

PAPER PRESENTATION

Optimization of the network and it's lookup Mar 2020 – Apr 2021

- Presented a paper to optimize the Internet and it's address lookup algorithm to meet the current and future needs by using Multipath Transmission Control Protocol (MPTCP), a better lookup for ipv6 address and by utilizing Segment Routing for Traffic Engineering